

Universität Freiburg – Institut für Physik

Applied Theoretical Physics

Prof. Dr. Michael Moseler

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Computational Physics: Materials Science - Exercise 2, SS 2026

Email: andreas.doell@physik.uni-freiburg.de

Tutorials: Mr. Andreas Doell

Problem 2. Dimir dynamics with the Morse potential

In this exercise, we study the motion of a singular H₂ dimir pair in a Morse potential, given by equation (1).

Be creative in this exercise! You need to think about your parameter choices and what **you** want to show with your simulation. Put your results into the context of the Morse potential and think about what features of the potential you observe in your simulations. The Morse potential is defined as

$$V(R) = D_e \cdot \left(1 - e^{(-a \cdot (R - R_e))}\right)^2, \quad (1)$$

where D_e is the so-called dissociation energy, R_e is the equilibrium distance of the dimir pair, R is its actual instantaneous distance and a controls the width of the potential.

Implementation

Implement a Python simulation which simulates such a system with Velocity-Verlet update rules. Start with an H₂ pair at a distance $r_e = R_e + \delta$, with $\delta \ll 1$ with a vanishing initial velocity $v_0 = 0$. Choose appropriate parameters and timestep Δt for your simulation.

- Check the conservation of the total energy $E=T+V$. What happens when you play around with your timestep or switch to Euler update rules?
- Familiarize yourself with the visualization software OVITO <https://www.ovito.org/> and the input format required for it to read your data. Write out the trajectory of your simulation into a file which is readable by OVITO and visualize it, in addition to plotting $x(t)$. At what frequency ω do the particles oscillate?
- Based on your initial tests and choice of parameters, decide on how many steps you need to run the simulation. For how long are you running your simulation?

Analysis

Convince yourself of the validity of your simulation and continue with the following analysis:

- Establish a relationship between the Morse potential and the known harmonic potential $V_H = \frac{k}{2}(x - x_0)^2$, with stiffness k . How can you test this in your simulation?
- Illustrate phase space spanned by (x, v) . How does it compare to the harmonic potential? Can you find interesting limits?
- What frequencies do you obtain for different r_e for $v_0 = 0$? Can you compare the function $\omega(r_e)$ to an analytical expression?

Play around with parameters and your simulation setup and observe what happens. Can you determine additional scaling laws?

What to submit

Hand in a zip file on ILIAS containing the following files:

- Python source code.
- A short report in pdf-format with your main results.

A good submission contains the following:

- A short description of the code and the goal of the exercise.
- A brief description how you managed to accomplish the task. This includes the choice of parameters and everything necessary to reproduce your exact plots.
- A *selection* of plots which display interesting features of the Morse potential with an explanation of what you observe.